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Optimizing agri-food supply chain configuration for mitigating both supply- and demand-side risks

Duxian Nie¹ and Haitao Li^{2*}

Abstract

Agri-food supply chains are increasingly vulnerable to disruptions from climate change, geopolitical conflicts, and pandemics, which manifest as both supply-side (e.g., volatile lead times) and demand-side uncertainties, threatening profitability and food security. While existing research has addressed these risks in isolation, a critical gap exists in tactically configuring perishable food supply chains to jointly mitigate both types of risk through integrated decision-making on sourcing, logistics, and inventory. This paper develops a novel Food Supply Chain Configuration Problem under Supply–Demand Risks (FSCCP-SDR). It is the first optimization model to simultaneously determine operational mode selection (supplier, transport, processing) and multi-echelon inventory placement (safety stock and early arrival stock) for perishable products under both stochastic demand and stochastic lead time, maximizing total net profit. To solve this NP-hard problem, we propose an exact dynamic programming method for small instances and a hybrid Particle Swarm Optimization-Genetic Algorithm (PSO-GA) with a custom repair mechanism for scalability. Applied to a grape processor case study in China, the hybrid PSO-GA finds near-optimal solutions (within 0.57% of optimal) efficiently. The model yields non-intuitive, data-driven configurations, such as selecting low-cost, high-loss options upstream and prioritizing reliability downstream. Sensitivity analyses reveal: (1) increased supply-side risk elevates both safety and early arrival stock, justifying investments in supplier reliability; (2) extended shelf-time significantly reduces required inventory, quantifying the value of preservation technologies; and (3) under extreme perishability, the optimal strategy pivots from "buffering with inventory" to "buffering with responsiveness." The methodology demonstrates strong scalability on larger synthetic networks.

Keywords Supply chain configuration, Agri-food supply chain, Supply risks, Stochastic lead time, Mixed-integer nonlinear programming, Genetic algorithm, Particle swarm optimization

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Introduction

The optimization of critical infrastructure systems under uncertainty is a paramount challenge across multiple domains, driven by global pressures such as climate change, geopolitical instability, and technological disruption. In the energy sector, for instance, significant research focuses on coordinating multi-energy hubs [1], managing decentralized grids with high prosumer penetration [2], and enhancing system resilience through hierarchical scheduling and demand response [3]. These studies underscore a universal theme: building economic and resilient performance in complex networks requires integrated models that can configure operational modes and manage buffers under profound uncertainty.

In parallel, agri-food supply chains—a vital global infrastructure—face analogous yet distinct challenges. While foods and agricultural products are necessities of our society, food insecurity continues to be a grand challenge of humanity in the twenty-first century [4]. On one hand, growing global population with industrialization and economic development are increasing the burden of feeding the world, especially the disadvantaged populations; on the other hand, a myriad of environmental issues such as shortage of arable land, drought, and flood have known to create significant risks for food systems [5], especially on the production and supply side. A recent study shows that nearly one third of global food crop production could be forced beyond the suitable conditions in the next six decades [6]. Other supply disruptions can be caused by pandemics of animal or human diseases. In 2019, African swine fever caused severe shortage of pork in China, which has a significant impact on pork-related products and supply, resulting in significant loss of GDP and sharp market price fluctuations in pork-related products globally [7]. The COVID-19 pandemic led to shortage of labor and shutdowns of meat processors, which forced farms to kill hundreds of thousands of pigs without processing them into food (NY [8]). The recent military conflict in Ukraine disrupted world's wheat exporter, causing global shortage of wheat supply and soaring prices [9]. These create unprecedented supply-side risks and disruptions.

Coping with supply risks and potential disruption calls for a resilient agri-food supply chain with two general strategies: sourcing and safety stock. While the traditional supply chain network design emphasizes cost efficiency and lean, such design can be quite prone to disruption without considering supply risks. How can one make better sourcing decisions while considering perceived risks of suppliers? Another well-known strategy is to increase the safety stock level, which works for either up fluctuation of demand or disruption of supply. Putting this in the context of a food supply chain, the question is which ingredients, components, or semi-finished

products in the corresponding bill-of-materials (BOM) should keep more safety stock. This goes beyond the traditional inventory control solution and practice that focus merely on managing inventories in one facility, e.g., processing plant or warehouse. Furthermore, the sourcing and safety stock decisions can interact with each other. For example, an overseas vendor is less costly but has longer lead time, which might result in more safety stock; a quality supplier can be more expensive but more reliable with less risk, which might in less safety stock. Thus one opts to address the sourcing and safety stock decisions simultaneously.

One way to approach to this is through the application of the supply chain configuration approach, which optimizes the mode selection and system-wide safety stock placement at the tactical level (cf. [10, 11]). The supply chain configuration decision can be made more frequently than the typical strategic supply chain network design decision [12], but less frequently than scheduling and routing at the operational level (cf. [13, 14]). For instance, proactive sourcing and inventory positioning planning can improve supplier/vendor selection and safety stock level by better balancing cost, lead time and quality [11]. In addition, supply chain configuration must be addressed while explicitly accounting for various supply-side uncertainties (cf. [15, 16]), which are ubiquitous in the food system [17, 18]. That is, how would anticipated shortages and prolonged transportation time impact supply chain performance? And given the forecasted supply risk, should sourcing and safety stock decisions be changed, and if so, how?

This has motivated our study of a new analytical approach for optimizing the configuration of food supply chains with both demand-side and supply-side risks. Inspired by advances in optimization and resilience modeling across interdependent systems (e.g., energy, water), our approach builds upon the classical supply chain configuration (SCC, [10]) framework and the existing model for food supply chain configuration of Li et al. [11], which simultaneously determines the mode selection (e.g., supplier/vendor selection, transportation mode) and safety stock placement decision to minimize the system-wide total supply chain costs. While most of the existing work on SCC considers merely the demand-side risk, our current work develops a more general food supply chain configuration model to explicitly account for both the demand-side and the supply-side risks, which is called the FSCCP-SDR in the sequel.

A review of the literature reveals two convergent limitations. First, strategic and resilience-focused models for agri-food chains (e.g., [19], [20]) do not address the tactical configuration problem of mode selection and inventory placement within a given network. Second, and more critically, the tactical Supply Chain Configuration

(SCC) methodology itself—including its food-specific adaptation by Li et al. [11]—fails to incorporate continuous supply-side process risk (stochastic lead time) alongside perishability and demand uncertainty. Consequently, no existing model provides a decision-support tool for optimizing the end-to-end configuration of a perishable food supply chain under simultaneous supply and demand risks. This gap hinders companies from making integrated, cost-effective resilience decisions at the tactical planning level.

To bridge this gap, this paper proposes the Food Supply Chain Configuration Problem under Supply–Demand Risks (FSCCP-SDR). Our work directly addresses the aforementioned shortcomings by: (i) integrating the stochastic lead time framework of Humair et al. [21] into the tactical SCC problem, and (ii) extending the perishable food SCC model of Li et al. [11] to explicitly account for supply-side variability. This creates the first unified modeling framework that allows joint optimization of mode selection, service time commitments, and multi-echelon buffer stocks under a single profit-maximization objective.

An exact dynamic programming (DP) algorithm is applied to obtain optimal solutions for the FSCCP-SDR, which is limited to small-size instances due to NP-hardness of the problem. To obtain effective and efficient solutions to instances with reasonably large size, we devise and implement a hybrid algorithm based on particle swarm optimization (PSO) and genetic algorithm (GA) with custom designed encoding scheme and operators. As a case study, we apply our FSCCP-SDR model and algorithms on a grape supply chain at large food processor for raisins and grape products in China. A comprehensive computational experiment is performed to examine the optimal food supply chain configuration and performance metrics with respect to changes of key input parameters.

Our paper makes significant contributions to both the theory and practice of agri-food supply chain management, which can be summarized in three key perspectives.

Theoretical & Modeling Contribution: We propose the first integrated optimization model (FSCCP-SDR) for the tactical configuration of perishable food supply chains under simultaneous supply and demand risks. This work bridges two previously disconnected streams of literature: (i) the stochastic lead time inventory theory [21] and (ii) the perishable food supply chain configuration model [11]. By making supply-side process risk (via stochastic lead time) an endogenous decision variable alongside mode choice and inventory placement, the FSCCP-SDR provides a unified analytical framework to quantify the complex trade-offs between cost, lead time

variability, perishability, and service level—a critical capability absent in prior models.

Methodological & Algorithmic Contribution: To solve the resulting NP-hard MINLP problem, we design and implement a novel hybrid PSO-GA algorithm. Its core innovation is a problem-specific Check-and-Repair Algorithm (CRA) that enables effective search in the mixed discrete–continuous solution space by repairing infeasible offspring during crossover and mutation. This algorithm not only finds high-quality, near-optimal solutions efficiently for real-world cases but also demonstrates superior scalability and robustness on larger synthetic networks compared to standalone PSO or GA, advancing the metaheuristic toolkit for complex supply chain optimization.

Practical & Managerial Contribution: Through an in-depth case study and systematic sensitivity analysis, we translate model outputs into actionable managerial insights. We move beyond generic recommendations to reveal non-intuitive, data-driven strategies, such as accepting higher loss rates upstream while prioritizing reliability downstream, and the strategic pivot from inventory buffering to responsiveness buffering under extreme perishability. The model provides a quantitative decision-support tool that enables managers to evaluate the financial impact of investing in packaging, cold chain, supplier development, and information technologies, thereby linking operational resilience directly to economic performance.

The remainder of this paper is organized as follows. Sect. "[Literature Review](#)" reviews the related research literature. Sect. "[Optimization Model](#)" formally describes the FSCCP-SDR with its mixed-integer nonlinear programming (MINLP) formulation. The hybrid algorithm based on PSO and GA is described in Sect. "[Solution Methodology and Approaches](#)". Computational experiment and results are presented and discussed in Sect. "[Computational Study](#)". Managerial implications and recommendations are obtained in Sect. "[Managerial Implications and Recommendations](#)". Sect. "[Conclusions and Future Research](#)" draws conclusions and discusses future research opportunities.

Literature review

Given the vast literature of the relevant research, we structure our review into three streams (summarized in Table 1) to clarify our positioning: (1) strategic agri-food supply chain design, (2) tactical supply chain configuration (SCC), and (3) recent advances in resilience, uncertainty, and solution methods relevant to our integrative approach.

Table 1 Positioning of the literature and the research gap addressed by FSCCP-SDR

Research stream	Key focus & representative works	Primary limitations relative to this study
Strategic agri-food SC design	Network structure, facility location, capacity planning under uncertainty, sustainability. [12, 22–24]	Focuses on long-term, strategic network design. Does not solve the tactical problem of optimal mode selection and inventory placement within a given network structure
Recent resilience & climate-aware models	Integrating climate-driven variability, resilience metrics, and sustainability in agri-food systems. [19], [20], [25–28]	Primarily address strategic design or policy/regulatory levels. While critical for context, they do not provide a model for the tactical configuration of operational modes and multi-echelon inventories under continuous supply process risk
Tactical supply chain configuration (SCC)	Classic/Deterministic SCC: Joint optimization of mode selection & safety stock [10, 29, 30] SCC with Specific Uncertainties/Risks: Demand uncertainty (inherent in GSM), dual sourcing for disruptions [31], carbon emission [32]. Food SCC with Perishability: Li et al. [11] is seminal but assumes deterministic lead time	Lacks consideration of supply-side uncertainty (assumes deterministic lead time) No existing SCC model integrates stochastic lead times (continuous supply risk) with perishability (loss/quality) and mode selection. The model of Li et al. [11] neglects supply-side risk
Metaheuristics for SC Risk	Solving sourcing, inventory, and design problems under supplier disruption or uncertainty. [33, 34]	These methods are not applied to the specific problem of tactical configuration of perishable chains under simultaneous stochastic lead time and demand
This study (FSCCP-SDR)	Jointly optimizes mode choice, service-time commitments (ST^{in}, ST^{out}), and multi-echelon buffers (safety & early arrival stock) for a perishable chain under simultaneous stochastic demand and stochastic lead time to maximize profit	Fills the identified integrated gap

Strategic design and recent resilience models in agri-food chains

A vast body of work addresses the strategic design of agri-food networks, focusing on facility location, capacity sizing, and technology selection to optimize cost, service, and increasingly, sustainability objectives [22], [35]. Recent advances robustly incorporate demand uncertainty [36], disruption risks [37], and multi-objective trade-offs [23, 24]. However, these models operate at the strategic level, determining the network's physical and contractual backbone. Once such a network is established, a distinct tactical-level problem arises: configuring its operational details by choosing specific options at each node and setting inventory parameters. This tactical problem is the focus of our paper and is grounded in the Supply Chain Configuration literature.

Tactical supply chain configuration (SCC) and its evolution

The SCC problem, formalized by Graves and Willems [10], provides the methodological foundation for our work. It addresses the tactical planning of an acyclic supply chain by simultaneously selecting a processing/transportation mode from available options at each stage and determining the inbound/outbound service times to optimize system-wide cost, subject to a guaranteed service level.

Foundational and deterministic SCC models

The canonical SCC model and its early extensions primarily considered deterministic lead time, focusing on integrating decisions like platform product design [29], resource constraints [38], and multi-sourcing [39]. The Guaranteed Service Model (GSM) framework ensured feasibility for demand uncertainty through safety stocks, but supply side was considered reliable. Recent deterministic extensions have integrated green considerations [32] and multi-period planning [40]. The critical limitation of this line of work is the omission of supply-side process variability.

Advancing SCC to incorporate uncertainty

Recognizing this gap, researchers have begun to integrate various uncertainties into SCC. Some works have incorporated stochastic demand more explicitly or considered dual sourcing for disruption mitigation [31]. A significant step was taken by Humair et al. [21], who extended the GSM to incorporate stochastic lead time, providing the analytical basis for calculating safety stock under supply process variability. However, their model did not integrate mode selection, treating the supply chain structure as fixed. Most pertinent to our context is the work of Li et al. [11], which developed the first SCC model explicitly for food supply chains, incorporating perishability through loss rates and quality decay. This model bridges

SCC and agri-food logistics. Nevertheless, it retains the deterministic lead time assumption of classic SCC, thus failing to capture the pervasive supply-side risks in agriculture. A review of recent literature [41] confirms that no SCC model, whether general or food-specific, has yet integrated the stochastic lead time framework of Humair et al. [21] with the mode selection and perishability features of Li et al. [11]. This constitutes a fundamental research gap.

Solution methods and metaheuristics for risk

SCC problems are NP-hard, solved exactly via DP for small instances [10] or with metaheuristics like PSO and GA for scalability [42]. Hybrid metaheuristics are gaining traction for complex problems [34]. Separately, metaheuristics have been applied to optimize sourcing and inventory under supplier disruption risk [33]. However, a hybrid algorithm tailored for the mixed-integer nonlinear structure of a perishable food SCC problem with simultaneous supply–demand risks remains unexplored.

Synthesis and research gap

As synthesized in Table 1, existing research leaves a clear gap. Strategic and resilience models do not address tactical configuration. Existing tactical SCC models either ignore perishability or, crucially, assume away supply process risk (deterministic lead time). Therefore, the FSCCP-SDR model makes a distinct contribution: it is the first to jointly optimize supplier/mode choice, guaranteed service times, and both safety stock and early arrival stock in a profit-maximizing framework for perishable food chains, under simultaneous stochastic demand and stochastic lead time. This integrates and extends the lines of work by Humair et al. [21] and Li et al. [11]. To solve this challenging problem, we contribute a novel hybrid PSO-GA algorithm with a custom Check-and-Repair mechanism, advancing the methodological toolkit for risk-aware agri-food supply chain optimization.

Optimization model

This section formally presents the FSCCP-SDR, the modeling framework developed to fill the identified research gap. The model is designed to jointly optimize the two key tactical levers—mode selection and inventory positioning—while explicitly accounting for the three defining characteristics of high-risk agri-food chains: demand uncertainty, supply process uncertainty (via stochastic lead time), and product perishability (via loss rates and quality decay). By integrating these elements into a single profit-maximization objective, the model enables the evaluation of trade-offs that were previously unaddressed in a unified manner.

The FSCCP-SDR can be formally described as follows. Consider a general food supply chain consisting of a set

of stages (nodes) including production, procurement, processing, and distribution of multiple end products, which is represented by a graph $G(V, E)$. The node set V denotes entities (materials, ingredients, by-products, and end products) and functions (sourcing, production, transportation) in the supply chain, and the arc set E represents the dependent demand between pairs of nodes. Let coefficient ρ_{ij} on arc $(i, j) \in E$ indicate the amount of i needed for one unit output of j . Following Li et al. [11], each stage can be fulfilled through multiple alternative option/modes that differ in cost, lead time and loss rate of agri-food processing. End products set is denoted by F , external market demand of each $f \in F \subset V$ obeys a normal distribution with known mean μ_f and standard deviation σ_f . Let $W(i)$ be the set of stages that immediately succeed stage i , then the demand μ_i and standard deviation σ_i of stage i can be computed as:

$$\mu_i = \sum_{j \in W(i)} \rho_{ij} \cdot \mu_j \quad (1)$$

$$\sigma_i = \sqrt{\sum_{j \in W(i)} (\rho_{ij})^2 (\sigma_j)^2} \quad (2)$$

The lead time of each option is uncertain and assumed to follow normal distribution with known mean and standard deviation, which reflects the supply-side risk. Other assumptions of the model with service-guaranteed inventory positioning follow those in [10].

The FSCCP-SDR aims to determine the selection of options/modes for each node, and the corresponding inbound and outbound service times, to maximize the system-wide total net profit of the supply chain. An MINLP formulation for the FSCCP-SDR can be written as below.

Sets and Parameters

V : set of supply chain nodes.

S : set of initial (starting) nodes in the network.

F : set of nodes representing the end products, $F \subset V$

B : set of by-products $B \subset V$

M_i : set of available options for each node $i \in V$

$W(i)$: set of immediate successors of node $i \in V$

$U(i)$: set of immediate predecessors of node $i \in V$

c_{io} : direct cost added of option $o \in M_i$ for node $i \in V$

α_i : z-value of the corresponding service level for node $i \in V$

h_i : stock holding cost rate per time unit of node $i \in V$

q_i : the quality score of a node $i \in V$

η_b : amount of by-product generated by one unit of $b \in B$

e_i : the maximum shelf time of node $i \in V$

N : the number of nodes $N = |V|$

H : a scalar for the company's planning horizon.

μ_i : mean of external demand of end node $i \in F$

σ_i^d : standard deviation of internal demand of node $i \in V/F$

ρ_{ij} : multiplier between a pair of nodes (i, j) in the bill-of-materials (BOM), indicating the amount of i needed for one unit of j .

λ_{io} : rate of loss of option $o \in M_i$ for node $i \in V$

e_{io} : shelf time of option $o \in M_i$ for node $i \in V$

μ_{io}^L : expected value of the stochastic lead time of option $o \in M_i$ for node $i \in V$

σ_{io}^L : standard deviation of the stochastic lead time of option $o \in M_i$ for node $i \in V$

μ_i^d : gross mean demand of node $i \in V/F$. Following Li et al. [11], the gross mean demand of a node factoring in the corresponding loss rate λ_i is:

$$\mu_i^d = \frac{1}{1 - \lambda_i} \mu_i \quad (3)$$

While the rate of loss quantifies the effect of perishability on the quantity of entities, an alternative measure is needed to capture the change of quality through the supply chain. Following Li et al. [11], a nonlinear quality function is assumed to be:

$$q_i = q_i^0 \cdot e^{-\tau \mu_i^L / e_i} \quad (4)$$

where $0 \leq q_i \leq 1$, with 1 meaning the highest quality with respect to the benchmark; and 0 meaning the lowest quality with complete loss of value; q_i^0 is the initial (input) quality score of node i , and τ is the decaying factor.

We define the following decision variables to prescribe the main decisions on the selection of options/modes and the inbound/outbound service times in the MINLP model of the FSCCP-SDR problem.

$$x_{io} = \begin{cases} 1 & \text{if option } o \in M_i \text{ is selected for node } i \in V \\ 0 & \text{otherwise} \end{cases}$$

$ST_i^{in} \geq 0$ as the inbound service time of node $i \in V$

$ST_i^{out} \geq 0$ as the outbound service time of node $i \in V$

The following additional derived decision variables are defined for the convenience of describing our MINLP formulation.

μ_i^L : expected value of the stochastic lead time of node $i \in V$

σ_i^L : standard deviation of the stochastic lead time of node $i \in V$

$C_i \geq 0$: direct cost added of node $i \in V$

$\lambda_i \geq 0$: rate of loss of each node $i \in V$

$\bar{C}_i \geq 0$: cumulative cost of node $i \in V$

$TT_i \geq 0$: net replenishment time of node $i \in V$

p_f : selling price per unit of end product $f \in F$

p_b : selling price per unit of by-product $b \in B$

Following Humair et al. [21], we first compute three statistics of the stochastic net-replenishment time $TT_i = \max(ST_i^{out} - ST_i^{in}, 0)$ of node i , i.e., $A^1(TT_i)$, $A^2(TT_i)$, and $A^3(TT_i)$ as below:

$$A^1(TT_i) = \int_{l=0}^{TT_i} f_i^L(l) dl \quad (5)$$

$$A^2(TT_i) = \int_{l=TT_i}^{\infty} l f_i^L(l) dl \quad (6)$$

$$A^3(TT_i) = \int_{l=TT_i}^{\infty} l^2 f_i^L(l) dl \quad (7)$$

Note that the density function $f_i^L(l)$ of node i 's lead time l is dependent upon the binary option selection decision variable x_{io} . That is, once option $o \in M_i$ is selected for node i , $f_i^L(l)$ can be characterized by the mean μ_i^L and standard deviation σ_i^L for normally distributed stochastic lead time considered in our study. Next, the nonnegative mean $R(TT_i)$ of TT_i and the nonnegative variance $J(TT_i)$ of the net replenishment time TT_i can be computed as:

$$R(TT_i) = A^2(TT_i) - TT_i(1 - A^1(TT_i)) \quad (8)$$

$$\begin{aligned} J(TT_i) &= TT_i^2 A^1(TT_i) (1 - A^1(TT_i)) \\ &\quad - 2TT_i A^1(TT_i) A^2(TT_i) \\ &\quad + A^3(TT_i) - (A^2(TT_i))^2 \end{aligned} \quad (9)$$

We then obtain the safety stock (SS_i), early arrival stock (EAS_i) and on-hand stock (OHS_i) of node i as below. We refer to Humair et al. [21] for the detailed derivation of these functions.

$$SS_i = \alpha_i \sqrt{R(TT_i)(\sigma_i^d)^2 + J(TT_i)(\mu_i^d)^2} \quad (10)$$

$$EAS_i = \mu_i^d(R(TT_i) - \mu_i^L + TT_i) \quad (11)$$

$$OHS_i = EAS_i + SS_i \quad (12)$$

Objective function

The objective of the FSCCP-SDR is to maximize the system-wide Total Net Profit (TNP), defined as the

difference between the Total Revenue (TR) and the Total Supply Chain Cost (TSC), as shown in Eq. (13).

$$\text{Maximize } TNP = TR - TSC \quad (13)$$

Total Revenue (TR): The revenue generated from selling end products and by-products is not constant but is a function of the quality score (q_i) of the product at the point of sale. Quality deteriorates nonlinearly with cumulative lead time relative to shelf time (Eq. 4). Thus, TR is calculated as:

$$TR = \sum_{f \in F} p_f q_f \mu_f + \sum_{b \in B} p_b q_b \eta_b \mu_b \quad (14)$$

where p_f and p_b are the base selling prices per unit, μ_f and μ_b are the mean demands, and η_b is the yield factor for by-product b . This formulation explicitly links operational decisions (which affect lead time and thus quality) to financial outcomes.

Total supply chain cost (TSC)

This is the sum of all costs incurred across the supply chain and consists of three core components (i.e., on-hand safety stock cost ($OHSC$), pipeline stock cost (PSC), and cost of goods sold ($COGS$)), integrating both demand and supply uncertainties:

$$TSC = \sum_{i \in V} h_i \bar{C}_i [\alpha_i \sqrt{R(TT_i) (\sigma_i^d)^2 + J(TT_i) (\mu_i^d)^2} + \mu_i^d (R(TT_i) - \mu^{L_i} + TT_i)] + \sum_{i \in V} h_i (\bar{C}_i - \frac{C_i}{2}) R(TT_i) \mu_i^d + \sum_{i \in V} HC_i \mu_i^d \quad (15)$$

A detailed breakdown of the TSC components is as follows:

(1) On-Hand Safety Stock Cost ($OHSC$):

$\sum_{i \in V} h_i \bar{C}_i [\alpha_i \sqrt{R(TT_i) (\sigma_i^d)^2 + J(TT_i) (\mu_i^d)^2} + \mu_i^d (R(TT_i) - \mu^{L_i} + TT_i)]$, this term captures the cost of holding the safety stock (SS_i) and the early arrival stock (EAS_i) that are physically present at a node i . The expressions inside the brackets correspond directly to Eqs. (10) and (11). The holding cost is applied to the cumulative cost ($\bar{C}_i$) of the product at that node, representing the capital tied up in inventory.

Safety	Stock	Component:
$\alpha_i \sqrt{R(TT_i) (\sigma_i^d)^2 + J(TT_i) (\mu_i^d)^2}$		calculates the required buffer against uncertainty. $R(TT_i)$ and $J(TT_i)$

are the mean and variance of the net replenishment time, translating lead time variability into an inventory requirement.

Early Arrival Stock Component: $\mu_i^d (R(TT_i) - \mu^{L_i} + TT_i)$ represents inventory that arrives earlier than the worst-case planned time. While not a planned buffer, it incurs holding costs and is influenced by the distribution of lead time.

(2) Pipeline Stock Cost

$(PSC): \sum_{i \in V} h_i (\bar{C}_i - \frac{C_i}{2}) R(TT_i) \mu_i^d$, this represents the cost of holding inventory that is in transit or being processed (i.e., work-in-process). The amount of pipeline stock is proportional to the mean demand and the average replenishment time $R(TT_i)$. The cost is applied to the average value of the material in the pipeline, approximated as $(\bar{C}_i - \frac{C_i}{2})$.

(3) Cost of Goods Sold ($COGS$): $\sum_{i \in V} HC_i \mu_i^d$,

this is the direct, variable cost of production and procurement incurred to fulfill the mean gross demand μ_i^d . It sums the direct cost C_i of the selected option at each node across the entire chain, effectively representing the base cost of goods before any inventory-related costs.

Interplay within the Objective: The objective function (13) creates a direct trade-off. Decisions that reduce costs (e.g., choosing a low-cost but slow/unreliable mode) may increase lead time (μ^{L_i}, σ_i^L) and loss rates (λ_i), which in turn can increase inventory costs ($OHSC$, PSC) and decrease revenue (TR) through quality decay and higher required demand fulfillment (μ_i^d). The FSCCP-SDR model optimally balances these intertwined effects to find the configuration that maximizes overall profitability under risk.

Constraints

The formulation (13) includes the following constraints.

$$\sum_{o \in M_i} x_{io} = 1, \forall i \in V \quad (16)$$

$$\mu_i^L = \sum_{o \in M_i} \mu_{io}^L x_{io}, \forall i \in V \quad (17)$$

$$\sigma_i^L = \sum_{o \in M_i} \sigma_{io}^L x_{io}, \forall i \in V \quad (18)$$

$$\lambda_i = \sum_{o \in M_i} \lambda_{io} x_{io}, \forall i \in V \quad (19)$$

$$C_i = \sum_{o \in M_i} c_{io} x_{io}, \forall i \in V \quad (20)$$

$$e_i = \sum_{o \in M_i} e_{io} x_{io}, \forall i \in V \quad (21)$$

$$\bar{C}_i = C_i + \sum_{j \in U(i)} \rho_{ji} \bar{C}_j, \forall i \in V \setminus S \quad (22)$$

$$ST_j^{in} \geq ST_i^{out}, \forall i \in V \setminus F, j \in W(i) \quad (23)$$

$$x_{io} \in \{0,1\}, ST_i^{in}, ST_i^{out} \geq 0 \quad (24)$$

$$\mu_i^L, \sigma_i^L, C_i, \lambda_i, \bar{C}_i, TT_i \geq 0 \quad (25)$$

The constraints (16)–(25) collectively ensure the feasibility and relevance of the FSCCP-SDR solution for perishable product supply chains. Their managerial logic is contextualized as follows:

- (1) Sourcing and Operational Mode Selection (Constraint 16): The requirement to select exactly one option per node ($\sum_{o \in M_i} x_{io} = 1$) is fundamental in the agri-food context for several reasons beyond mere model simplification. It enforces a definitive and manageable sourcing strategy at each stage. For perishable products, this is critical because different options (e.g., different suppliers, processing methods, or transport modes) are characterized by distinct loss rates (λ_{io}), quality decay profiles (linked to shelf time e_{io} and lead time), and cost structures (c_{io}). *Committing to a single option per node allows for:*
- (2) Predictable Quality and Loss Management: It enables the model to precisely calculate the cumulative product loss and quality deterioration along the chain, which is essential for accurate revenue estimation (via Eq. 13) and cost accounting.
- (3) Operational Consistency: It ensures that the selected safety stock and early arrival stock levels are calculated based on a consistent set of parameters (lead time, loss rate), preventing the operational chaos and unpredictability that would arise from frequently switching between disparate modes for the same product flow.

This constraint models a common tactical planning practice where a primary mode is established for a planning period, creating a stable baseline configuration against which risks are mitigated through inventory positioning.

- (2) Parameter Propagation and Cumulative Cost (Constraints 17–22): Constraints (17)–(21)

propagate the attributes (mean lead time μ_i^L , lead time deviation σ_i^L , loss rate λ_i , shelf time e_i , direct cost C_i) of the selected option to the node. This is essential for perishables because quality decay and loss are cumulative and path-dependent. The cumulative cost $\bar{C}_i$ (Constraint 22) reflects the total cost invested in a product unit by the time it reaches node i , which directly determines the financial value at risk when holding inventory (affecting holding cost $h_i * \bar{C}_i$ in the objective function). Accurate cumulative cost is vital for correctly valuing inventory of perishable goods whose value erodes over time.

- (3) Service Time Consistency and Shelf-time Feasibility (Constraint 23): The constraint $ST_j^{in} \geq ST_i^{out}$ ensures that a downstream stage j cannot promise to deliver to its customer (its outbound service time) before it has guaranteed to receive inputs from its upstream stage i . In an agri-food chain, this has a critical implication for perishability. The total elapsed time from the initial node to the final customer, influenced by these service times and the selected lead times, must be implicitly less than the effective shelf time of the product to avoid spoilage. While not an explicit constraint, the model enforces this through the objective function: configurations leading to excessive total lead times that exceed shelf time will result in catastrophic loss ($\lambda_i \rightarrow 1$) and near-zero quality ($q_i \rightarrow 0$), rendering them non-optimal due to plummeting revenue (Eq. 13) and soaring effective costs. Thus, the optimal solution naturally respects shelf-time limits.
- (4) Domain and Non-negativity (Constraints 24–25): These define the solution space, ensuring binary selection decisions and non-negative times and costs, which are standard for physical and economic realism.

Operational definition of resilience in the FSCCP-SDR framework

In this paper, we adopt an operational, performance-based definition of supply chain resilience within the context of our optimization model. Resilience is not modeled as an explicit constraint or a separate objective function. Instead, it is an emergent property of the optimally configured system that manifests through the model's endogenous decisions when facing uncertainty. Specifically:

We define resilience as the system's capacity to maintain service levels and limit the collapse of total net profit (TNP) in the face of increased supply-side (σ_{io}^L) and demand-side (σ_f) variability. This capacity is enabled and measured by the model's ability to optimally reallocate resources, which includes:

- (1) Dynamically adjusting buffer stocks: Rebalancing the levels and positions of safety stock (to hedge against demand and lead time uncertainty) and early-arrival stock (to buffer against lead time variability).
- (2) Endogenously switching operational modes: Selecting more reliable (often lower σ_{io}^L) but potentially costlier options at critical nodes when the perceived risk justifies the investment.

The objective of maximizing TNP (Eq. 13) inherently drives these resilience-building actions. A configuration that fails to adequately adapt to risk will incur higher penalty costs (e.g., lost sales from stockouts, or excessive inventory costs), resulting in a lower TNP. Therefore, the profit-maximizing solution simultaneously identifies the most economically efficient and, by our definition, the most resilient configuration for a given risk profile. This approach aligns with the business reality where investments in resilience must ultimately be justified by their contribution to the financial viability and stability of the operation. The sensitivity analyses in Sect. "Sensitivity Analysis" will explicitly illustrate this dynamic interplay between risk, configuration decisions, and system performance (resilience).

Discussion on model assumptions and robustness

The FSCCP-SDR model assumes that both external demand and processing/transportation lead times follow normal distributions. This assumption is common in foundational inventory and supply chain configuration literature (e.g., [10, 21]) as it yields tractable, closed-form expressions for safety stock (Eq. 10) and early arrival stock (Eq. 11) within the GSM framework. It provides a crucial first-order approximation of uncertainty, capturing the essential trade-offs using mean and variance.

Limitations and contextual justification We fully acknowledge that real-world agri-food supply chain uncertainties can deviate significantly from normality. Supply-side disruptions (e.g., port closures, extreme weather) can cause lead times to exhibit highly skewed or heavy-tailed distributions, such as the log-normal distribution. Similarly, demand shocks (e.g., panic buying) can create non-normal patterns. The normality assumption is a deliberate simplification adopted in this initial, integrated model of supply–demand risk for food SCC. Its purpose is to establish a parsimonious and solvable analytical foundation that clearly elucidates the joint impact of both risk dimensions on configuration decisions, without the overwhelming complexity of fitting and integrating arbitrary, non-normal distributions at each node.

Robustness and extensibility of the framework

The core contribution of our work—the integrated modeling framework and the hybrid solution algorithm—is not inextricably tied to the normality assumption. The PSO-GA metaheuristic operates on the objective function value and constraints; it is agnostic to the underlying probability distributions. Therefore, the FSCCP-SDR framework is inherently adaptable. Future research could extend it in several ways to enhance realism:

- (1) Incorporating Non-Normal Distributions: The model could be adapted to use log-normal, Gamma, or Weibull distributions for lead times, which are better suited for modeling positive, skewed durations. This would require deriving or approximating new expressions for the statistics $R(TT_i)$ and $J(TT_i)$ (Eqs. 8–9) or employing numerical integration.
- (2) Simulation-Based Optimization (Sim-Opt): For empirically observed or complex distributions, a simulation module could replace the analytical inventory calculations. The hybrid PSO-GA would then evaluate candidate configurations via simulation, searching for robust solutions—a powerful albeit computationally intensive direction.
- (3) Distributionally Robust Optimization (DRO): If only moments (mean, variance) or support sets are known, a DRO approach could be integrated to find configurations that perform well under the worst-case distribution within an ambiguity set.

These avenues represent important future work to broaden the model's applicability. The current formulation serves as the necessary and rigorous stepping stone, providing the structural insights and algorithmic platform upon which these more complex, distributionally-aware models can be built.

Solution methodology and approaches

The FSCCP-SDR is a generalization of the deterministic SCC, which is well-known to be NP-hard [42]. This classification in computational complexity theory indicates that the time required to find an optimal solution grows exponentially with problem size in the worst case, making exact methods impractical for large instances.

Overall solution strategy

Given the NP-hard nature of the FSCCP-SDR, we adopt a two-tiered solution strategy:

- (1) Exact Solution for Validation: For small-scale instances, we implement an exact Dynamic Programming (DP) algorithm, adapted from the framework of Graves and Willems [10] and Humair

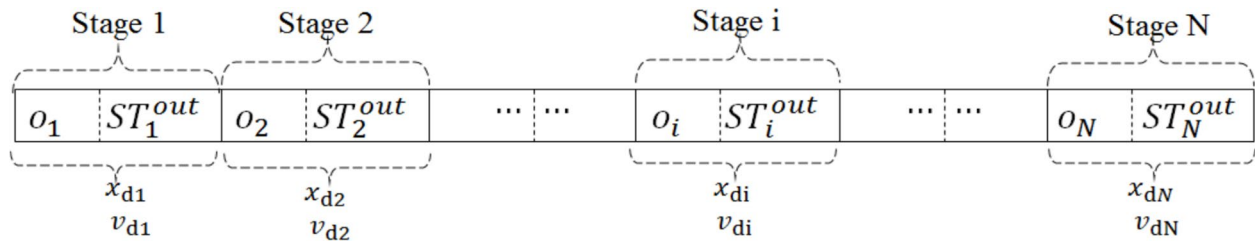


Fig. 1 Schema of each particle

If $i \in S$ then:

$$ST_i^{in} = 0; ST_i^{out} = \mu^{L_i} * rand();$$

Else If $i \in V/F$:

$$ST_i^{in} = \max(ST_j^{out}), j \in U(i); ST_i^{out} = (ST_i^{in} + \mu^{L_i}) * rand();$$

Else:

$$ST_i^{in} = \max(ST_j^{out}), j \in U(i)$$

$$ST_i^{out} = 0; i \in F$$

End If

Fig. 2 Pseudo code of the heuristic algorithm (HA)

and Willems [43] to incorporate stochastic lead time. The complete DP formulation is provided in the supplementary material. This serves as a benchmark to validate the quality of heuristic solutions.

- (2) Metaheuristic Solution for Scalability: For realistically sized problems, we develop a novel hybrid Particle Swarm Optimization (PSO, Kennedy and Eberhart [44]) and genetic algorithm (GA, [45]), called PSO-GA. Core methodological innovations of our approach lie in: (1) a problem-specific solution representation that directly encodes both discrete mode selections and continuous service times, (2) a hybrid PSO-GA framework that leverages PSO's efficient global search and GA's operators for diversification, and most critically, (3) a customized Check-and-Repair Algorithm (CRA) designed to handle the unique feasibility challenges arising from the interaction between discrete and continuous decision variables during crossover and mutation. These components are detailed below, which allows us to efficiently navigate the mixed discrete–continuous solution space of FSCCP-SDR.

Hybrid PSO-GA algorithm

This subsection details the design of our primary solution algorithm. Figure 3 provides the high-level pseudo-code.

Solution representation and initialization

The position x_d and velocity v_d of each particle X_d can be simply expressed as $(o_1, ST_1^{out}, o_2, ST_2^{out}, \dots, o_N, ST_N^{out})$, as shown in Fig. 1,

with a length of $2N$, where o_i is the selected option for node i (discrete integer) and ST_i^{out} is its outbound service time (continuous), N is the number of nodes in the supply chain network. That is,

$$x_d = (x_{d1}, x_{d2}, \dots, x_{dN}) = (o_1, ST_1^{out}, o_2, ST_2^{out}, \dots, o_N, ST_N^{out})$$

$$v_d = (v_{d1}, v_{d2}, \dots, v_{dN}) = (o_1, ST_1^{out}, o_2, ST_2^{out}, \dots, o_N, ST_N^{out})$$

The selected alternative option o_i and the service time ST_i^{out} are respectively initialized as follows: $o_i = |M_i| * rand()$. The service time ST_i^{out} is initialized by using a heuristic algorithm (HA) below, as shown in Fig. 2. $rand()$ is a function that generates a random number uniformly distributed in $[0, 1]$.

Core PSO-GA and the check-and-repair algorithm (CRA)

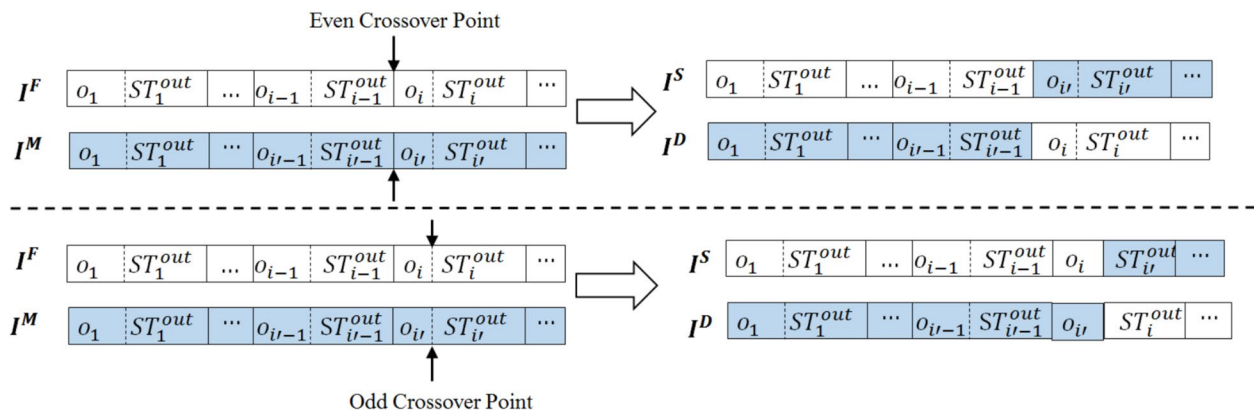
Our hybrid PSO-GA algorithm can be conceptually described by Fig. 3.

The hybrid algorithm executes PSO's velocity-position initialization and update (Steps 1–4) followed by GA-inspired crossover and mutation (Steps 5–6). The crucial innovation is applied during these GA steps to handle the feasibility issue inherent in our representation.

Crossover and the Feasibility Challenge: During crossover (Step 5), two parent particles I^F and I^M exchange genetic material at a randomly selected crossover point, as shown in Fig. 4. A significant problem arises if this point falls on an odd index (i.e., within a pair (o_i, ST_i^{out})). In this case, an offspring may inherit an option o_i from one parent but the corresponding ST_i^{out} from the other. This decoupled (o_i, ST_i^{out}) pair may violate the precedence constraint $ST_j^{in} \geq ST_i^{out}$ in Eq. (23), because the inherited ST_i^{out} was calculated based on the lead time of the other parent's option, not its own.

The Check-and-Repair Algorithm (CRA): To resolve this, we designed the dedicated CRA, as shown in Fig. 5. For each node in an infeasible offspring, CRA checks if the inherited inbound service time ST_i^{in} is consistent with the newly

Step 1: Initialize parameters and the particle swarm $X = \{X_1, X_2, \dots, X_d, \dots, X_n\}$ with random positions and velocities.
Step 2: Calculate and evaluate the fitness of each particle.
Step 3: Update the position and velocity of each particle.
Step 4: Compare the fitness of each particle with its best position, that is, compare all current $pbest$ with $gbest$, update $gbest$.
Step 5: Perform crossover operator , that is, select m (a specified ratio) of particles from the particle swarm size n for random pairwise crossover to generate the same number of offspring particles. Call check-and-repair algorithm (CRA) to repair infeasible offspring particles.
Step 6: Perform mutation operator , that is, randomly select a fragment of the particle individual and let it mutate with the probability p_m . Call CRA to repair infeasible the mutation particle individual.
Step 7: Check whether the hybrid algorithm meets stopping criteria. If it does, output the optimal solution results. Otherwise, go to Step 2 .

Fig. 3 Pseudo code of the hybrid PSO-GA algorithm**Fig. 4** Crossover with even or odd crossover point

assigned option o_i . If inconsistent, it recalculates a feasible ST_i^{in} (and consequently ST_i^{out}) within the solution space defined by the new option's lead time, using the heuristic initialization logic. This repair mechanism is what enables the use of standard crossover operators on our complex, mixed-variable representation.

Mutation (Step 6) faces a similar issue. The CRA is invoked if mutation alters an o_i to ensure the ST_i^{out} remains feasible for the new option.

The detailed execution process steps of the hybrid PSO-GA algorithm are described as follows:

In Step 1, to initialize the particle swarm with random positions and velocities.

In Step 2, the fitness of all particles is evaluated by objective function (13). The best position and fitness of the particle in the particle individual extremum is denoted by $pbest$, and the best position and optimal fitness in the global extremum is denoted by $gbest$.

For I^S and I^D

ST_i^{in} or ST_i^{in} will be checked to determine whether it is in new solution space defined by the fragment of the alternative option o_i or o_i ;

If ST_i^{in} or ST_i^{in} is not in new solution space

A new ST_i^{in} or ST_i^{in} will be created randomly in its new solution space;

Else

Nothing will be done.

End If

End For

Fig. 5 Pseudo code of the checking and repairing algorithm (CRA)

In Step 3, use (26) and (27) to update the position x_d and velocity v_d of each particle:

$$x_{di}^{k+1} = x_{di}^k + v_{di}^{k+1} \quad (26)$$

$$v_{di}^{k+1} = wv_{di}^k + c_1r_1^k(pbest_{di}^k - x_{di}^k) + c_2r_2^k(gbest_i^k - x_{di}^k) \quad (27)$$

where w is an inertia weight, c_1 and c_2 are the learning factors, and r_1 and r_2 are random numbers between (0,1). k is the current number of iterations, i is the i -th of the stage, and d represents the d -th particle.

In Step 4, compare the fitness of each particle with its best position, and if it is similar, use the current position as the best position of the particle. Compare all current $pbest$ with $gbest$, update $gbest$.

In Step 5, select m particles from the particle swarm size n for random pairwise crossover to generate the same number of offspring particles. The selected two particles as parents are denoted as I^F and I^M , respectively, which creates two new offspring particles denoted as I^S and I^D with crossover probability p_c . As shown in Fig. 4, there can be two types of crossovers: even crossover point or odd crossover point, depending on whether there exists a particle i whose option o_i and the service time ST_i^{out} are separated. When a random crossover point occurs in an even number position of parents I^F and I^M , the offspring particle can readily inherits its parents' solutions. However, when the random crossover point occurs in an odd number position of parents, the offspring partially inherits o_i from its parent I^F , and partially inherits ST_i^{out} from its parent I^M , which can be infeasible. In this case, we devise a check-and-repair (CRA) algorithm as shown in Fig. 5. In Step 6, a particle is operated by mutation with the probability p_m . In the mutation process, we randomly select a fragment of a particle individual

and replace its value by a random value. Similarly, we also consider different cases where the fragment mutation position is an odd or even number because it may produce an infeasible particle individual: (i) If the mutation point is an even number, the fragment of the service time ST_i^{out} is replaced by the gene ST_i^{out} reproduced using HA; (ii) If the mutation point is an odd number, the fragment of the alternative option o_i is replaced by the fragment o_i reproduced in its solution space, and the fragment of the service time ST_i^{out} need be checked to determine whether it is a feasible particle. If it is infeasible, apply CRA to restore feasibility.

Parameter tuning and validation

Performance of the hybrid PSO-GA is affected by several key parameters in the algorithm, i.e., crossover ratio, probability of crossover, and probability of mutation. In this section, at first, we use a one-factor-at-a-time (OFAT) approach to present some results on parameter tuning to achieve best average performance; then, we conducted a follow-up, more systematic analysis to validate the some results on parameter using fractional factorial design. For a particular instance of FSCCP-SDR in the case study to be presented in Sect. "Computational Study", Fig. 6 shows the best average objective values of 10 runs when the crossover ratio m varies in the set {0.1, 0.2, 0.3, 0.4, 0.5, 0.6}. While the average TNP fluctuates with m , $m = 0.5$ achieves the best performance.

We then change the probability p_c of the crossover operation in the set: {0.1, 0.3, 0.5, 0.7, 0.9}. As shown in Fig. 7, the best average total cost increases as p_c increases from 0.1 to 0.3, then it decreases as p_c increases to 0.7 before it goes up again.

We next change the probability p_m of mutation in the set {0.00485, 0.0097, 0.0194, 0.0388, 0.0776}. Figure 8 shows that the average objective value can be improved by increasing p_m initially until it reaches 0.0194, before it goes up with further increase of p_m . This suggests that

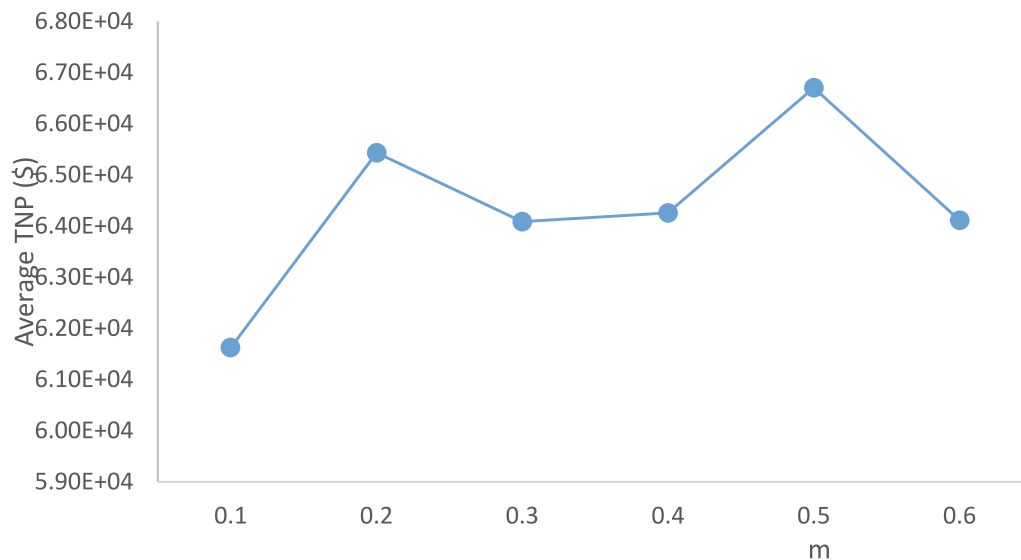


Fig. 6 Average total net profit with different crossover ratio m

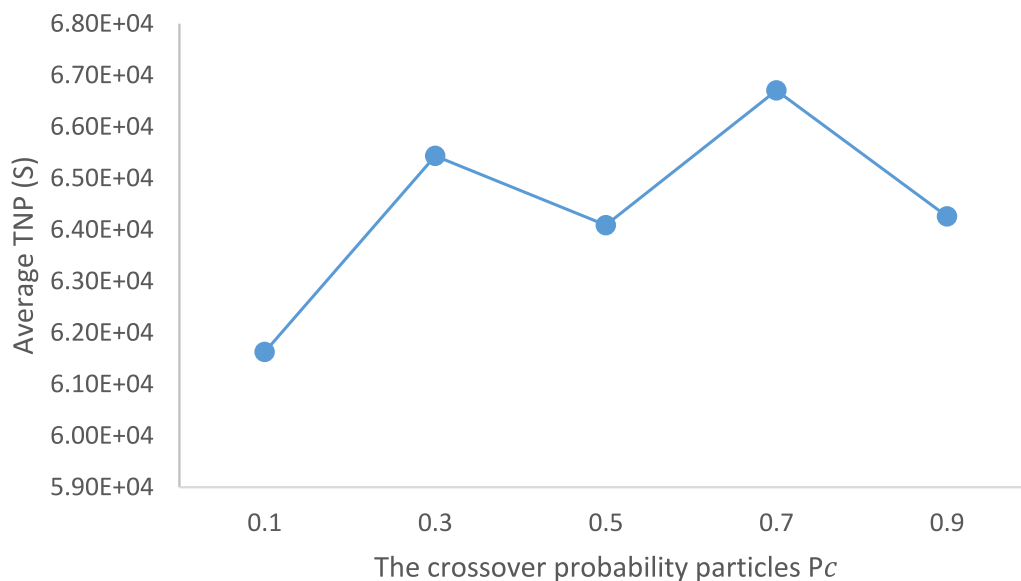


Fig. 7 Average total net profit with different probabilities of crossover operation

a moderate level of diversification via mutation tends to perform better.

To address the limitation of OFAT—its inability to detect parameter interactions—and to enhance the credibility of our parameter selection, we conducted a follow-up, more systematic analysis using a 2^k fractional factorial design. We focused on the three key parameters ($k = 3$), each at two levels (a "low" and a "high" value around the baseline identified by OFAT). The design allowed us to efficiently estimate the main effects and the most important two-way interactions of these parameters on the algorithm's performance (measured by the average TNP over 10 runs).

The analysis of variance (ANOVA) from this design confirmed that:

- (1) The main effects of p_c and p_m were statistically significant ($p < 0.05$), aligning with the trends observed in the preliminary analysis.
- (2) The interaction effect between p_c and p_m was not statistically significant for the tested instance, suggesting that the OFAT approach did not miss a critical synergy or conflict between these parameters within the studied region.
- (3) The previously identified baseline parameter combination ($m = 0.5$, $p_c = 0.7$, $p_m = 0.0194$) performed consistently well within the design space

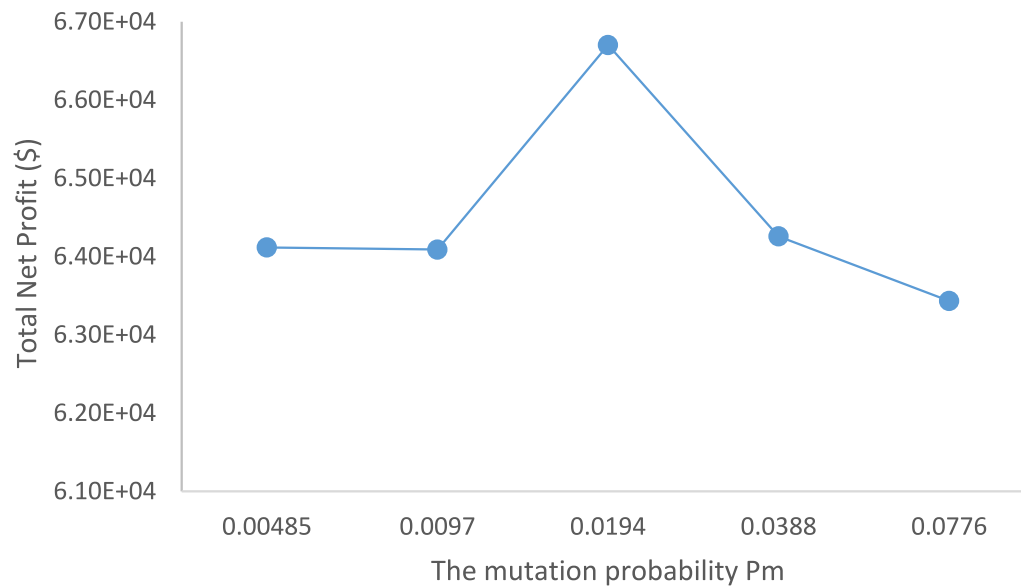


Fig. 8 Average total net profit with different probabilities of mutation

and was not statistically outperformed by other tested combinations.

This combined approach—initial OFAT screening followed by fractional factorial validation—provides a balanced and rigorous justification for the parameter set used in our comparative experiments. While more exhaustive methods like the Taguchi method, full Response Surface Methodology (RSM), or meta-optimization could potentially fine-tune parameters further, they are computationally prohibitive given the expense of each FSCCP-SDR evaluation. Our two-stage process ensures that the reported performance of the hybrid PSO-GA is based on a reasonably optimized and validated parameter set, not an arbitrary one.

Thus we set m , p_c and p_m to be 0.5, 0.7 and 0.0194, respectively, in our PSO, GA and hybrid PSO-GA algorithms for the computational study.

Implementation details for reproducibility

To ensure the reproducibility of our research, we specify the following implementation details:

- **Programming and Environment:** All algorithms (DP, PSO, GA, Hybrid PSO-GA) were coded in MATLAB R2018b.
- **Solver Use:** The MINLP model for fitness evaluation within the metaheuristics was implemented natively in MATLAB. No external commercial solver was used for the overall configuration optimization to maintain flexibility within the heuristic search loops.
- **Hardware Platform:** All computational experiments were performed on a standard PC with an Intel

(R) Core (TM) i5-7200U CPU @ 2.50 GHz processor and 4 GB RAM, running the Windows 10 operating system.

- **Key Algorithmic Parameters:** For the hybrid PSO-GA, the key parameters were set as: population size $n = 200$, maximum iterations = 100, inertia weight $w = 0.8$, learning factors $c_1 = c_2 = 2$, crossover ratio $m = 0.5$, crossover probability $p_c = 0.7$, and mutation probability $p_m = 0.0194$. The random seed was controlled for the benchmarking experiments in Sect. "A Case Study".
- **Data Availability:** The data used in the case study are described in Sect. "A Case Study". Synthetic instances for scalability testing were generated using a script that is available from the corresponding author upon reasonable request.

Computational study

We apply our model and algorithms on a case study of a grape processor in China to demonstrate the decision-support provided by the FSCCP-SDR. A comprehensive computational experiment is also conducted to further examine the impacts of various input parameters on the optimal FSCCP-SDR solutions.

A case study

Data Sources, Composition, and Preparation:

The input data for this case study is synthesized from multiple sources to ensure realism and relevance:

- (1) **Structural & Baseline Operational Data:** The supply chain network structure (Fig. 9) and baseline parameters for direct costs, nominal shelf time, and

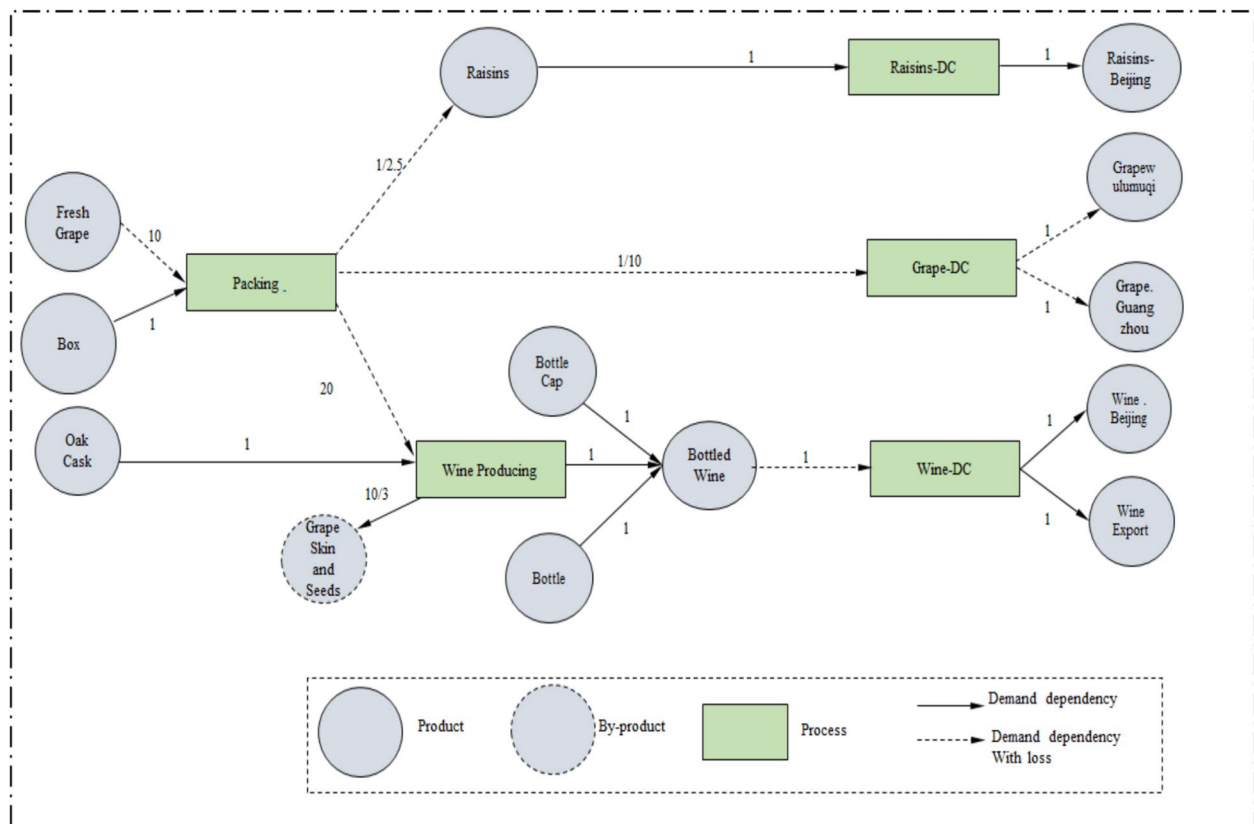


Fig. 9 An example of grape supply chain network (adapted from [11])

loss rates for various processing/packaging options are adapted from the validated case study of Li et al. [11]. This provides an academically grounded foundation.

- (2) Augmented Supply-Side Risk Data: To model supply risk—the core extension of this study—we augmented the dataset by introducing stochastic lead time. The mean lead times (μ_{io}^L) were set equal to the deterministic values from Li et al. [11]. The standard deviations (σ_{io}^L) were estimated based on expert elicitation from industry practitioners and literature benchmarks on agricultural logistics variability, resulting in coefficients of variation between 0.25 and 0.5.
- (3) Market & Financial Data: Selling prices and demand statistics (mean μ_f , standard deviation σ_f) for end-products were derived from a combination of company-provided estimates, regional market reports, and scaling based on typical firm size. All monetary values were adjusted to a consistent base year using relevant indices.
- (4) General Parameters: Service level (z-value, corresponding to α_i), inventory holding cost rate (h_i), and planning horizon (H) are set based on standard industry practice and company policy.

All data has been reviewed for internal consistency and face validity by a domain expert. The complete, consolidated dataset used for the computational experiments is presented in Table 1. The following sections detail the company's decision questions and our analysis.

Consider a medium-size grape processor, Morning Star Company, in Wulumuqi, Xinjiang Province, China, adapted from Li et al. [11]. Its main products include fresh grapes, raisins and wine, which are sold to Beijing, Guangzhou and exported overseas to Europe and Africa. Due to severe climate change and fierce market competition in recent years, Morning Star has been experiencing significant supply chain challenges. For instance, some farmers and producers found it harder to predict the grape yields given the irregular precipitation pattern, which in turn made the lead time of order fulfillment fluctuate more. Having observed increased variation of lead times from suppliers by the purchasing department of Morning Star, the operations manager would like to address the following questions:

- Should the amount of safety stock and early arrival stock be increased in response to more variation of lead time? And if so, where and how much, i.e. how would the safety stock and early arrival stock placement be?

Table 2 Input data for the grape supply chain case study (sources: adapted from Li et al. [11], augmented with expert-estimated stochastic parameters)

Stage	Measurement	Options	Direct cost c_{io} (\$) [Type: Continuous]	Shelf life e_{io} (days) [Type: Continuous]	Loss rate λ_{io} [Type: Continuous, 0–1]	lead time (<i>Mean ± Standard Deviation, days</i>) [Type: continuous]		Data source key
						Mean μ_{io}^L	Standard deviation σ_{io}^L	
Fresh grape	kilograms(kg)	Manual	2.0	2	0.03	2	0.5	L+E
		Machine	4.2	4	0.01	1	0.5	L+E
Box	units	Regular	7.0	–	0.0	1	0.5	L+E
		Cushioned	10.0	–	0.0	3	0.5	L+E
Oak cask	units	In-house	18.0	–	0.0	15	1	L+E
		Buy	30	–	0.0	5	0.5	L+E
Packing	boxes	Regular	0.50	10	0.04	1	0.5	L+E
		Iced	0.8	20	0.02	2	0.5	L+E
Raisins	kg	Natural dry	0.65	180	0.05	50	1.5	L+E
		Machine dry	0.96	250	0.03	20	1	L+E
Wine producing	kg	Procedure-1	45	720	0.02	30	1.5	L+E
		Procedure-2	60	900	0.01	15	0.5	L+E
Bottle cap	units	Sichuan	7	–	0.0	4	0.5	L+E
		Local	10	–	0.0	2	0.5	L+E
Bottle	units	Sichuan	6	–	0.0	4	0.5	L+E
		Local	9	–	0.0	2	0.5	L+E
Bottled wine	bottles	Standard	10	720	0.02	5	1	L+E
		Deluxe	14	800	0.01	2	0.5	L+E
Raisins-DC	kg	Regular	0.1	180	0.04	3	0.5	L+E
		Conditioned	0.25	300	0.02	1	0.5	L+E
Grape-DC	kg	Regular	0.1	4	0.04	3	0.5	L+E
		Conditioned	0.25	7	0.02	1	0.5	L+E
Wine-DC	kg	Regular	0.5	720	0.02	3	0.5	L+E
		Conditioned	1	800	0.01	1	0.5	L+E
Raisins-Beijing	kg	Train	0.25	180	0.04	4	0.5	L+E
		Truck	0.4	250	0.02	2	0.5	L+E
Grape-Wulumuqi	kg	Regular	0.03	4	0.04	2	0.5	L+E
		Conditioned	0.06	10	0.02	1	0.5	L+E
Grape-Guangzhou	kg	Regular	1	4	0.05	5	0.5	L+E
		Conditioned	1.5	10	0.03	3	0.5	L+E
Wine-Beijing	bottles	Train	0.25	720	0.02	7	0.5	L+E
		Truck	0.8	800	0.01	3	0.5	L+E
Wine -Export	bottles	Ocean	5.0	720	0.02	7	0.5	M+E
		Air	15	800	0.01	3	0.5	M+E

Standard deviation values represent supply-side risk. Data Source Key: L=Parameter adapted from Li et al. [11]. E=Stochastic parameter (standard deviation) estimated/validated via expert input and literature benchmarking. M=Derived from market data/company estimate. S=Standard industry or model parameter. “–” indicates the parameter is not applicable for that stage/option. All cost data are in US Dollars (USD). Time units are in days.

Table 3 The mean and standard deviation of demand of end product nodes (in tons)

Mean of demand	30	15	200	70	80
Standard deviation of demand	10	4	90	10	20
Selling price p_f (\$)	200	80	160	850	1200

- How would the change of maximum shelf time, loss rate, lead time and demand affect the optimal choice of options for each stage of an end-to-end rice supply chain, e.g., operational and transportation modes?

To answer these questions with data-driven decisions support, the Supply Chain Analytics team at Morning Star approaches to the decision problem as an FSCCP-SDR. Morning Star’s grape supply chain network can be depicted by Fig. 9, with 18 nodes including 12 entities (ingredients and parts), five processes and one by-product. Table 2 provides the direct cost, shelf time, loss rate, and mean and standard deviation of lead time for the available options of all nodes except the by-product node.

We set z-value to 1 corresponding of the service level α_i and the inventory holding cost rate h_i assuming 60%

Table 4 Comparison of solutions found the base scenario

Parameters	DP	PSO-GA	PSO	GA
<i>TNP</i> (\$)	67,085.57	66,703.42	64,435.93	66,603.89
<i>TR</i> (\$)	91,961.13	86,901.71	89,397.05	91,758.87
<i>TC</i> (\$)	24,875.56	20,198.29	24,961.12	25,154.98
<i>SSC</i> (\$)	223.68	310.60	222.63	323.63
<i>PSC</i> (\$)	1410.95	2285.58	1420.88	2908.08
<i>COGS</i> (\$)	23,226.95	17,598.43	23,307.77	21,906.03
<i>OHSC</i> (\$)	237.66	314.28	232.47	340.87
Safety stock level	4420.22	5342.25	4545.63	5138.64
Early arrival stock level	411.57	1055.37	439.70	
Computational time (seconds)	421.43	178.98	143.92	211.95
Optimality gap (%)	0.00%	0.57%	3.95%	0.72%
Improvement over PSO (%)	-	+ 3.52%	-	-
Improvement over GA (%)	-	+ 0.15%	-	-
Inference	Benchmark	Best balance: Near-optimal solution with reasonable time	Fastest but solution quality significantly lower	Good quality but slower than hybrid

1) Improvement over PSO (%) = $[(TNP_PSOGA - TNP_PSO) / TNP_PSO] \times 100\%$; Improvement over GA (%) = $[(TNP_PSOGA - TNP_GA) / TNP_GA] \times 100\%$.

2) Detailed breakdown of safety stock cost (SSC), pipeline stock cost (PSC), and other components is consistent with the model and shows the hybrid algorithm achieves lower total cost primarily through better configuration reducing COGS.

inventory holding cost annual rate is 0.0015 for each stage. Table 3 shows the selling price, mean and standard deviation of external market demand of end products.

Each instance is solved by the exact DP algorithm, PSO, GA and the hybrid PSO-GA algorithm. Quality of the PSO-GA solutions can be evaluated by the gap from the DP solutions: $\frac{|Obj - Obj_{DP}|}{Obj_{DP}}$, where Obj_{DP} represents the optimal objective value obtained by the DP and Obj is best objective value obtained by a metaheuristic algorithm.

The solutions to the case study by different algorithms are shown in Table 4. With moderately more computational time, the hybrid PSO-GA achieves the best quality with 0.57% gap, which outperforms GA with 0.72% gap and PSO within 3.95% gap. The hybrid PSO-GA achieves an improvement of approximately 3.52% in TNP over the standalone PSO and 0.15% over the standalone GA. Table 5 shows the optimal configuration solutions found by the DP algorithm. For the optimal options selected, we characterize them by: Low, Average, and High. The optimal configuration recommends a mixture of high-cost and low-cost options for the nodes, with tradeoffs between shelf time, loss rate and supply risk. It is interesting to note that the optimal configuration tends to recommend options of low cost and high loss rate for early stages of the supply chain, with a balanced level of supply risks throughout all stages of the supply chain. Such optimal configuration would not be intuitive without our optimization approach.

Configuration pattern summary

The optimal solution exhibits a clear strategy: select low-cost, high-loss-rate options at upstream stages (e.g., raw material processing) where losses are less costly to absorb, and shift towards more reliable (lower risk)

Table 5 Optimal configuration by the DP algorithm

	Nodes	o_i	ST_i^{out}	Direct cost	Shelf life	Loss rate	Supply risk
1	Fresh grape	1	1	Low	Low	High	Low
2	Box	1	1	Low	-	-	High
3	Oak cask	1	2	Low	-	-	Low
4	Packing	1	1	Low	High	High	Average
5	Raisins	1	33	Low	Low	High	Low
6	Wine producing	1	15	Low	Low	High	Low
7	Bottle Cap	2	0	High	-	-	High
8	Bottle	1	1	Low	-	-	Low
9	Bottled wine	1	7	Low	Low	High	Low
10	Raisins-DC	1	27	Low	Low	High	Average
11	Grape-DC	2	1	High	High	Low	High
12	Wine-DC	1	4	Low	Low	High	Low
13	Raisins-Beijing	1	0	Low	Low	High	Low
14	Grape-Wulumuqi	1	0	Low	Low	High	Low
15	Grape-Guangzhou	2	0	High	High	Low	High
16	Wine-Beijing	1	0	Low	Low	High	Low
17	Wine -Export	2	0	High	High	Low	High

Supply risk is characterized by the **standard deviation** of lead time, categorized as Low, Average, or High based on the selected option's lead time variability

options at downstream stages closer to the customer to ensure service level and preserve accumulated product value. This non-intuitive pattern highlights the value of system-wide optimization over local cost minimization.

Sensitivity analysis

In order to further examine the impacts of key parameters, a set of sensitivity analysis is performed. We focus on three key parameters: demand variation, stochastic lead time, loss rate, and shelf time.

Demand variation

We vary the external demand variance of the end node by multiplying the base σ_f by a coefficient in the set {0.5, 1.0, 1.5, 2.0}. Figure 10 illustrates the impact. As expected, total supply chain cost increases nonlinearly (Fig. 10(a)), and total net profit decreases significantly (Fig. 10(b)) with greater demand uncertainty.

The behavior of inventory components offers a deeper insight. Safety stock increases while early arrival stock decreases as demand variation grows (Figs. 10(c) and (d)). This opposing trend arises because demand variation directly affects the required safety stock coverage (Eq. (10)), while the model optimally re-allocates the buffer strategy between the two inventory types within the constraints of service times.

Managerial implications for agri-food practice

- **Investment in Demand Sensing & Forecasting:** The nonlinear cost increase with demand volatility (Fig. 10(a)) underscores the high economic value of improving demand forecast accuracy. Investments in advanced forecasting tools, point-of-sale data integration, and collaborative planning with retailers can reduce effective σ_f , yielding direct supply chain cost savings.
- **Dynamic Inventory Policy Reconfiguration:** The opposing trends in safety stock and early arrival stock (Figs. 10(c) & (d)) indicate that a static inventory policy is suboptimal. Managers should implement dynamic inventory rebalancing protocols. In periods of high demand uncertainty, the system should automatically lean towards increasing safety stock while potentially drawing down early arrival stock, and vice-versa, as guided by the model's logic.
- **Product and Market Portfolio Strategy:** The significant profit erosion under high demand variation (Fig. 10(b)) provides a quantitative basis for evaluating market segments. Products or customer channels with inherently stable demand profiles contribute disproportionately to profitability and resilience.

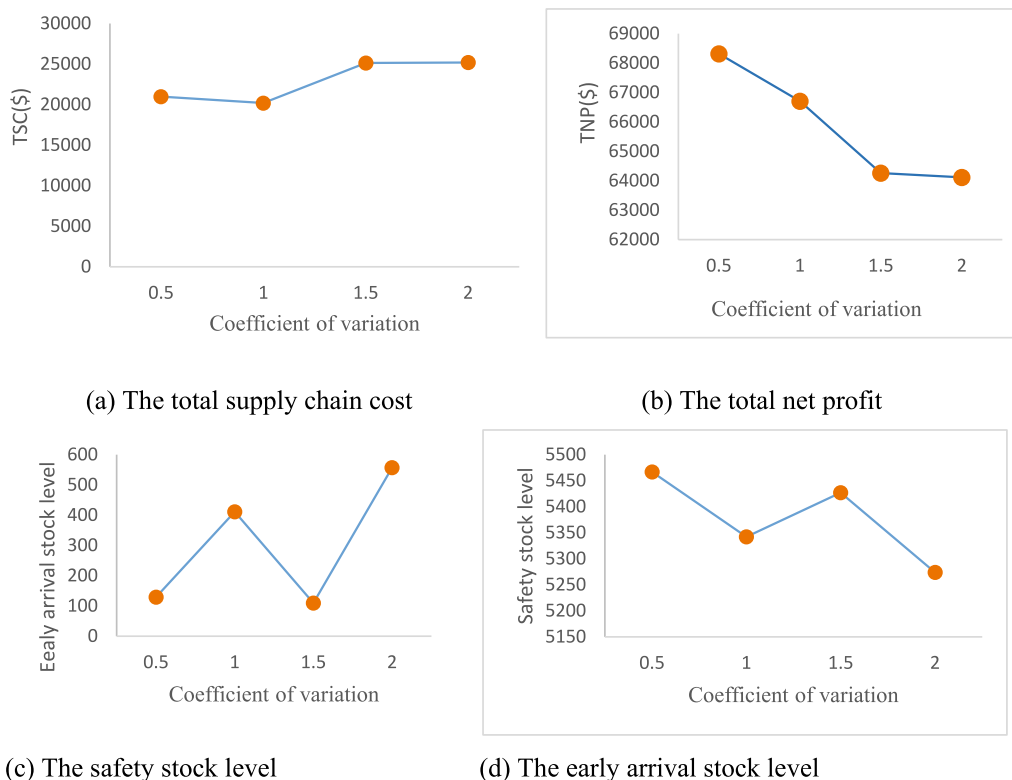


Fig. 10 Effect of demand variation on total net profit, total cost, safety stock and early arrival stock (a) The total supply chain cost (b) The total net profit. (c) The safety stock level (d) The early arrival stock level

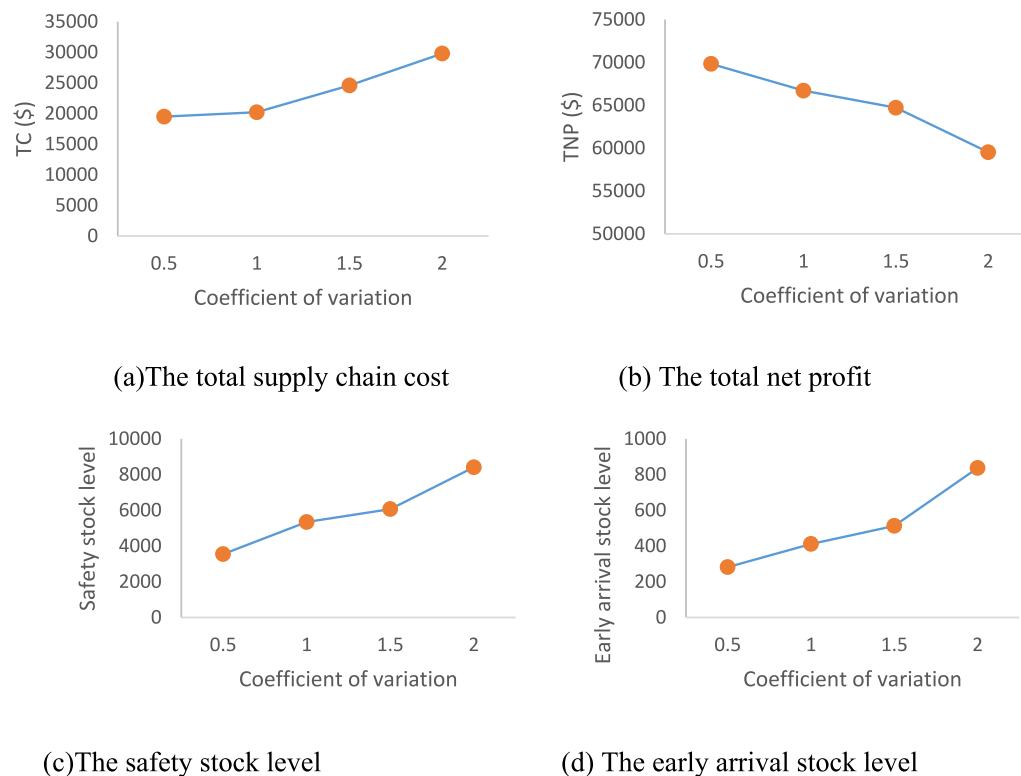


Fig. 11 Effect of the variance of lead time on total net profit, total cost, safety stock and early arrival stock. (a) The total supply chain cost (b) The total net profit (c) The safety stock level (d) The early arrival stock level

Supply-side risk

We vary the lead time variance of each stage by multiplying the base σ_{io}^L with a coefficient in the set {0.5, 1, 1.5, 2.0}. The total supply chain cost increases with larger lead time variation, i.e., higher supply-side risk, as in Fig. 11(a), and total net profit decreases with lead time variation as in Fig. 11(b). Both the safety stock and the early arrival stock increase with larger lead time variation as in Fig. 11(c) and Fig. 11(d).

Managerial implications for agri-food practice

- **Supplier Development & Performance Management:** Since lead time variation (σ_{io}^L) directly drives up both safety and pipeline inventory (Fig. 11(c)&(d)), companies should formally measure and benchmark supplier lead time reliability. Contracts can include incentives/penalties tied to this metric, and collaborative projects can focus on reducing variability at source (e.g., better harvest scheduling).
- **Logistics Technology Adoption:** Investments in real-time transportation visibility platforms and predictive analytics for logistics can reduce transit time uncertainty. The model allows quantifying the inventory cost savings from such investments, justifying their return on investment (ROI).

- **Dual-Sourcing Justification:** For stages where lead time variability is high and costly to mitigate (Fig. 11a), the model quantifies the "risk cost." This cost can be directly compared to the premium of procuring from a more reliable (but potentially more expensive) backup supplier, providing a clear economic rationale for multi-sourcing strategies.

Shelf time

We vary the shelf time of each stage by multiplying the base e_{io} with a coefficient in the set {0.5, 1, 1.5, 2.0}. Figure 12 shows that both the safety stock and the early arrival stock decrease significantly with shelf time variation, as shown in Fig. 12 (c) and (d), respectively. This means that longer shelf time, through food engineering, packaging and storage technologies, may significantly reduce safety stock.

Managerial implications for agri-food practice

- **Packaging Technology ROI Analysis:** The strong negative relationship between shelf time and required safety stock (Fig. 12a) turns packaging from a cost center into a strategic investment. The model enables a direct cost–benefit analysis for investing in modified atmosphere packaging, active packaging, or improved barrier materials. The reduction in

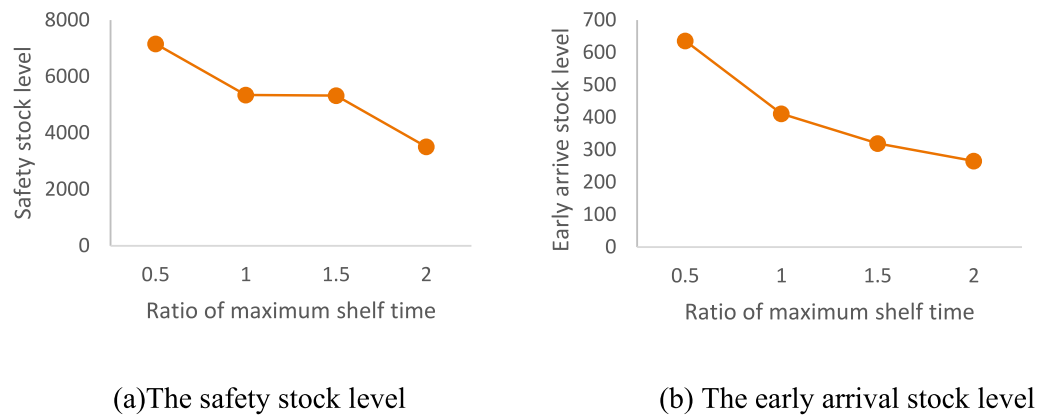


Fig. 12 Effect of maximum shelf time on safety stock and early arrival stock. (a) The safety stock level (b) The early arrival stock level

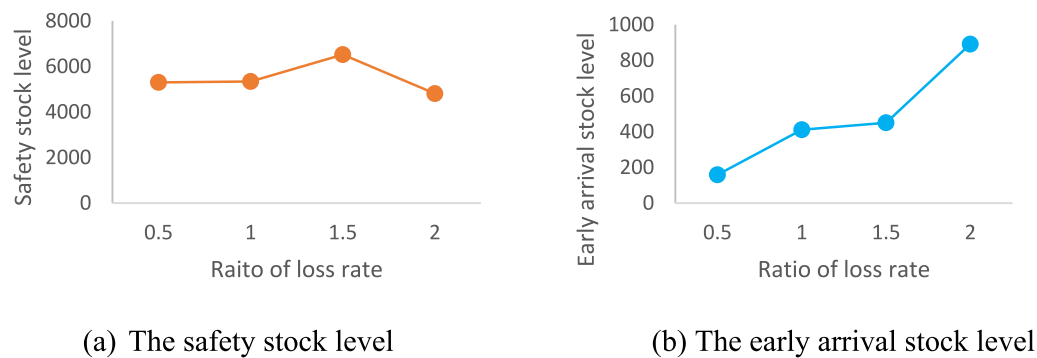


Fig. 13 Effect of loss rate on safety stock and early arrival stock. (a) The safety stock level (b) The early arrival stock level

inventory holding costs (capital tied up) can offset the higher unit packaging cost.

- **Cold Chain Infrastructure Justification:** For temperature-sensitive products, investments in cold chain integrity (e.g., better pre-cooling, refrigerated transport, consistent warehouse temperatures) effectively extend shelf time by slowing decay. Our results provide a quantitative link between cold chain quality (reflected in extended e_{io}) and reduced working capital requirements.
- **Process Innovation Priority Setting:** Processing steps that significantly extend shelf time (e.g., gentle drying, pasteurization) should be evaluated not only on their direct cost but also on their system-wide inventory reduction benefit, which is made visible by this analysis.

Loss rate

We now vary the loss rate of each stage by multiplying the base λ_{io} with a coefficient in the set {0.5, 1, 1.5, 2.0}. When loss rate increases, both safety stock and early arrival stock tend to increase as shown in Fig. 13(c) and Fig. 13(d). Note that when the loss rate is very high, the model may tradeoff safety stock with other cost

components of COGS and pipeline stock cost, such that the safety stock may go down.

Managerial implications for agri-food practice

- **Targeted Loss Reduction Initiatives:** The model helps prioritize loss reduction efforts. Stages with high loss rates (λ_{io}) that also feed into high-cumulative-cost downstream nodes have the greatest financial impact. Resources for process improvement, better handling equipment, or staff training should be allocated accordingly.
- **Trade-off Analysis for Sourcing:** A supplier offering a lower price but higher loss rate may not be cheaper overall. The model automatically internalizes the cost of the lost material and the inflated inventory needed to compensate for it (Fig. 13), enabling a true total cost of ownership comparison between suppliers.
- **Resilience through Redundancy vs. Inventory:** At very high loss rates, the model may suggest a counterintuitive decrease in safety stock, opting to manage risk through other means. This highlights a critical trade-off: when perishability is extreme, building redundant capacity or flexible rerouting

options might be more cost-effective than holding large stocks of highly perishable inventory.

Deep insight: strategic buffer transformation under extreme perishability

The sensitivity analysis on loss rate reveals a nuanced and powerful insight that goes beyond simple trend descriptions. As shown in Fig. 13, when loss rates become very high (coefficient > 1.5), the model's optimal response may not be to further increase safety stock. Instead, it may begin to reduce upstream safety stock levels while concurrently shifting the configuration toward faster, more reliable (and often more expensive) modes at downstream stages.

This behavior illustrates a fundamental strategic shift in buffer management identified by the model:

- (1) "Buffering with Inventory": Under moderate loss rates, holding safety stock is an effective buffer against variability.
- (2) "Buffering with Responsiveness": Under extreme perishability, holding inventory for long periods is financially equivalent to creating waste. The model recognizes that the cost of stock decay outweighs the benefit of safety coverage. Therefore, it abandons inventory as the primary buffer and seeks resilience through increased supply chain responsiveness—choosing modes with shorter, more reliable lead time to reduce the need for long-duration holding.

This insight is crucial for managers: "When product perishability dominates, the optimal risk mitigation strategy pivots from carrying more stock to investing in speed and reliability." The FSCCP-SDR model does not merely recommend "hold more stock when risk increases"; it

intelligently identifies the conditions under which to change the buffering strategy entirely, providing a sophisticated, context-aware decision-support tool.

Algorithm scalability and methodological generalizability

While the grape processor case study provides deep, context-specific managerial insights, a valid concern is whether the proposed FSCCP-SDR framework and solution methodology are specific to that particular network or possess broader applicability. To address this and demonstrate the structural generalizability and scalability of our approach, we evaluate its performance on a set of synthetically generated FSCCP-SDR instances with varying network sizes (40, 60, 80, 100, 120, and 160 nodes). None of these instances can be solved by the DP algorithm due to their size. Each instance is solved 10 times by the PSO, GA and the hybrid PSO-GA algorithms. The average total supply chain cost and computational time are shown in Fig. 14 and Fig. 15, respectively.

Purpose of synthetic instances

These instances are not tied to any specific product or company. They are constructed with randomized but realistic parameters (costs, lead times, loss rates) within plausible ranges for agri-food operations, following the general structure of a multi-echelon acyclic network. Their primary role is to test whether our model formulation and hybrid PSO-GA algorithm can effectively solve the FSCCP-SDR problem as its scale increases, independent of the case-specific data.

Results as evidence of generalizability

As shown in Fig. 14, the hybrid PSO-GA (hpg) algorithm consistently outperforms standalone PSO and GA across all problem sizes, and its relative advantage grows with

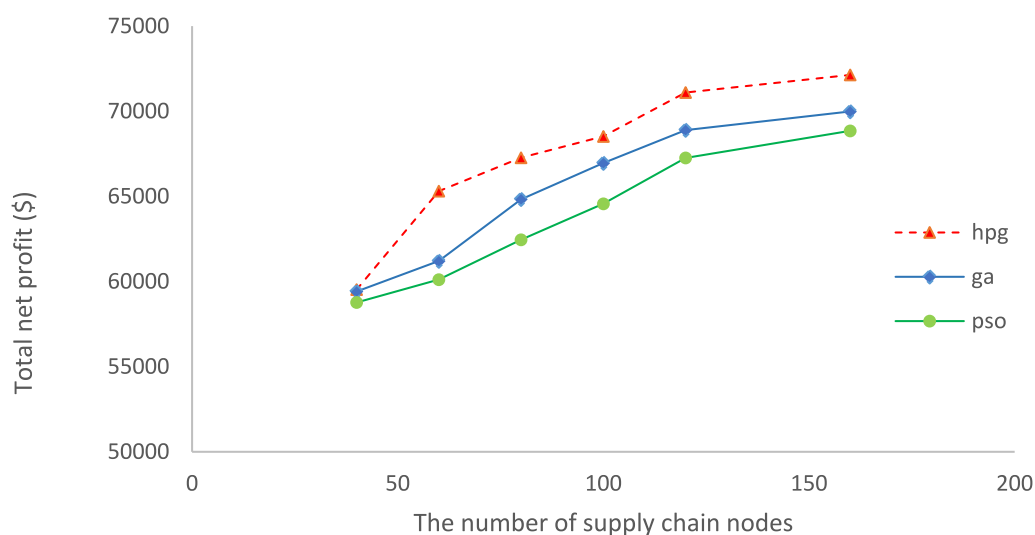


Fig. 14 Average net total profit for large-sized instances

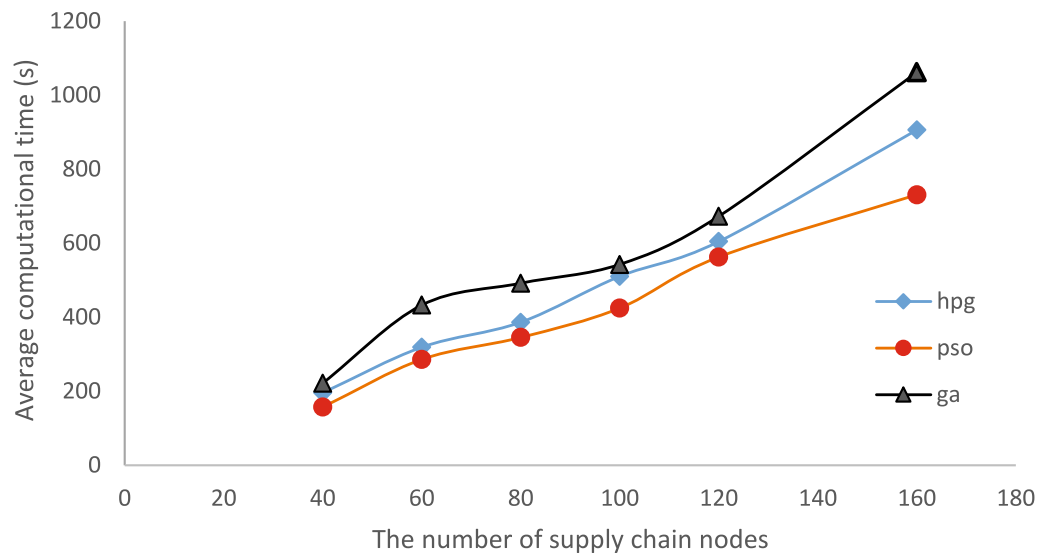


Fig. 15 Average time for large-sized instances (in seconds)

scale. Figure 15 shows it maintains a reasonable computational time. Crucially, this consistent performance across diverse, randomly generated network structures demonstrates that our solution methodology is robust and not tailored to the idiosyncrasies of the grape supply chain. It scales effectively to problems that are structurally more complex and larger than the original case.

Therefore, these computational experiments serve a dual purpose: (1) they benchmark the algorithmic performance, and (2) more importantly, they provide evidence that the core FSCCP-SDR modeling and solution approach is generalizable to tactical configuration problems in agri-food supply chains of varying sizes and compositions. The managerial insights derived from the case study should be interpreted in light of its specific parameters, but the optimization framework itself is portable.

The hybrid PSO-GA (hpg) algorithm outperforms GA and PSO for all the instances, and its advantage tends to increase when problem size increases. As shown in Fig. 14, the hybrid PSO-GA achieves a higher total net profit (TNP) across all problem scales. Quantitatively, across all large instances (40 to 160 nodes), the hybrid PSO-GA achieves an average improvement of approximately 4.2% in TNP over the standalone PSO and 1.8% over the standalone GA. This demonstrates the consistent and scalable superiority of the hybrid approach. As shown in Fig. 15, GA spends most computational time, followed by the hybrid PSO-GA, which is moderately more than PSO. The hybrid algorithm's additional computational time (compared to PSO) is justified by its significant solution quality improvement, as evidenced by the percentage gains in TNP.

Managerial implications and recommendations

The FSCCP-SDR model and the case study analysis yield actionable insights for managers responsible for agri-food supply chain planning and risk mitigation. We distill these into the following key recommendations:

- (1) Adopt an Integrated Risk Assessment Lens:
- (2) Move beyond demand-only planning. Proactively quantify and incorporate supply-side risk metrics (e.g., lead time variation from suppliers and logistics) into tactical configuration models alongside demand uncertainty.
- (3) Use a unified framework like FSCCP-SDR to evaluate the joint impact of supply and demand risks on total cost and service level, rather than managing them in silos.
- (4) Strategically Balance Mode Selection and Inventory Placement:
- (5) Avoid simplistic "lowest-cost" sourcing. The optimal configuration often involves a mix of high- and low-cost options across the network. Consider upstream nodes with lower direct cost but higher loss rates, balanced against downstream nodes where reliability and shorter lead times become critical to preserve product value and meet service commitments.
- (6) Jointly optimize safety stock and early arrival stock. Recognize that these are two distinct buffers against different uncertainties. The model provides a data-driven method to determine their optimal levels and locations, preventing costly over-investment in inventory.
- (7) Leverage Technology and Process Improvement to Create Configuration Flexibility:

- (8) Invest in shelf-time extension. Results indicate that improving shelf-time (via better packaging, cold chain, or processing) can significantly reduce required safety stock levels. The cost of such technologies can be directly evaluated against the resulting inventory carrying cost savings.
- (9) Evaluate information technologies that reduce lead time variability (e.g., real-time tracking, predictive analytics for logistics), as reducing supply-side risk (σ^L) lowers both inventory buffers and related costs.
- (10) Implement Dynamic Configuration Reviews:
- (11) Treat the supply chain configuration as a dynamic, tactical plan. Regularly re-run the configuration analysis (e.g., quarterly or semi-annually) in response to changes in key parameters: volatility of supplier lead times, shifts in market demand patterns, updates to loss rate estimates, or changes in sourcing options and costs.
- (12) Use sensitivity analysis (as demonstrated in Sect. "Sensitivity Analysis") to identify which parameters your network is most sensitive to, and prioritize monitoring and data collection efforts accordingly.
- (13) Build a Foundation for Advanced Strategies:
- (14) The single-sourcing optimal configurations identified by the model provide a strong baseline. Use this to strategically identify nodes where the cost of disruption is highest, thereby pinpointing candidate nodes for future multi-sourcing or backup supplier strategies, which would be a logical next-step investment in resilience.

Conclusions and future research

This study was motivated by a clear gap in the tactical planning literature for agri-food supply chains: the lack of an optimization model that simultaneously handles supply-side process risk (stochastic lead time), demand-side uncertainty, and product perishability in configuring operational modes and inventory buffers. The proposed FSCCP-SDR model was developed to bridge this specific gap. Our work demonstrates that by integrating the stochastic lead time analysis of Humair et al. [21] into the perishable-food SCC structure of Li et al. [11], we can create a practical tool for integrated risk mitigation. Key findings confirm that the model yields non-intuitive configurations and identifies strategic tipping points (e.g., from inventory-based to responsiveness-based buffering), precisely the kind of sophisticated insights that were inaccessible when supply risk was omitted. The hybrid PSO-GA algorithm ensures this framework is scalable.

While the managerial insights (Sect. "Managerial Implications and Recommendations") are derived from and calibrated to a specific case, the FSCCP-SDR optimization framework and the hybrid PSO-GA algorithm

are designed for general applicability. This is supported by their consistent and scalable performance on larger, synthetically generated networks with varying structures (Sect. "Algorithm Scalability and Methodological Generalizability"), confirming that the methodology is not limited to a single product or company context. Future research should, however, validate the model with data from different agri-food sectors to further enrich and generalize the practical insights.

Limitations and future research opportunities

This study has developed a tactical optimization framework (FSCCP-SDR) focused on economic resilience—maximizing profit while maintaining service levels under simultaneous supply and demand uncertainty. The model explicitly trades off cost, lead time variability, perishability (loss and quality), and inventory buffers. However, in line with the critical challenges motivating this work—including climate change—we acknowledge that a comprehensive assessment of agri-food supply chain configuration must eventually grapple with environmental and social sustainability dimensions.

The current formulation does not quantify environmental externalities such as carbon emissions, energy use, water footprint, or post-consumer waste. This is a deliberate scope limitation for this foundational model integrating supply-side risk, but it points to a vital avenue for extension. Future research should integrate such environmental metrics, drawing on methodologies from the growing literature on sustainable and circular agri-food supply chains (e.g., [23, 24]). This could involve:

- (1) Multi-objective optimization that balances economic profit against environmental impact (e.g., total carbon equivalent emissions).
- (2) Incorporating environmental regulations or cap-and-trade mechanisms as constraints or cost components within the FSCCP-SDR framework.
- (3) Extending the model attributes to include environmental performance indicators (e.g., emissions per unit for different transport or processing options), allowing the model to identify configurations that are both resilient and low-impact.

Exploring these integrations would align the tactical configuration decision with broader corporate sustainability goals and regulatory trends, making the model even more relevant for policymakers and industry leaders navigating the twin challenges of resilience and environmental stewardship.

Other future research directions include: (1) relaxing the single-sourcing assumption to consider multi-sourcing strategies; (2) relaxing the normality assumption to incorporate more realistic, skewed distributions (e.g.,

log-normal for lead times) via extended analytical forms or simulation–optimization, thereby enhancing model fidelity for severe disruption scenarios;(3) from a methodological perspective, applying advanced design of experiments (DOE) techniques, such as the Taguchi method or response surface methodology (RSM), to the metaheuristic parameter tuning process could further optimize algorithmic performance and robustness, especially for extremely large or complex problem instances.

In summary, the FSCCP-SDR addresses the critical shortcomings of prior SCC models by making supply-side risk endogenous to the configuration decision, thereby providing a more comprehensive foundation for building economically resilient agri-food supply chains.

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Author contribution

Dr. Haitao Li had the original conceptualization of the problem and modeling approach, and write the initial version of the manuscript. Dr. Duxian Nie implemented the model and solution methods. All authors reviewed the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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